

Quiz2 Answer

1. Answer the following questions according to the following **BOOKSTORE** relation:

Number	Name	City	Rank
1	Aaa	Taipei	20
2	Bbb	Taipei	15
3	Ccc	Hsinchu	10
4	Ddd	Taoyuan	20
5	Eee	Hsinchu	15
6	Fff	Taoyuan	10

(a) According to the above table, what attribute(s) can be used as a candidate key?

Answer: Number or Name can be used as a candidate key.

(b) Can (Number, Name) be a candidate key, why?

Answer: No, because (Number, Name) is not the simplest one, it cannot be a candidate key.

2. The table shown in the following lists dentist/patient appointment data. A patient is given an appointment at a specific time and date with a dentist located at a particular surgery (手術). On each day of patient appointments, a dentist is allocated to a specific surgery for that day.

StaffNo	DentistName	PatientNo	PatientName	Appoint-Date	Appoint-Time	SurgeryNo
S1011	Tony	P100	Gillian	06/09/2023	10:00	S12
S1011	Tony	P105	Jill	06/09/2023	12:00	S15
S1024	Helen	P108	Ian	06/09/2023	10:00	S10
S1024	Helen	P108	Ian	06/11/2023	14:00	S10
S1032	Robin	P105	Jill	06/11/2023	16:30	S15
S1032	Robin	P110	John	06/12/2023	18:00	S13

Suppose (**StaffNo, Appoint-Date, Appoint-Time**) is the **primary key**, and there are the following functional dependencies:

- (1) StaffNo → DentistName, (2) PatientNo → PatientName,
 (3) (StaffNo, Appoint-Date, Appoint-Time) → SurgeryNo

- (a) Describe what is **insertion anomaly** and show the insertion anomaly that this relation contains. (2%)
 (b) Describe what is **deletion anomaly** and show the deletion anomaly that this relation contains. (2%)
 (c) Describe what is **update anomaly** and show the update anomaly that this relation contains. (2%)
 (d) Describe and illustrate the process of normalizing the above table to 3NF relations. (You have to normalize it to 2NF relations first, then to 3NF relations.) (6%)

(d)

Answer: 2NF (Remove Partial Dependency)

(StaffNo, DentistName), (2NF, 3NF)

(StaffNo, PatientNo, PatientName, Appoint-Date, Appoint-Time, SurgeryNo) (2NF but not 3NF)

Then 3NF (**Remove Transitive Dependency**)

(PatientNo, PatientName) (3NF)

(StaffNo, PatientNo, Appoint-Date, Appoint-Time, SurgeryNo) (3NF)